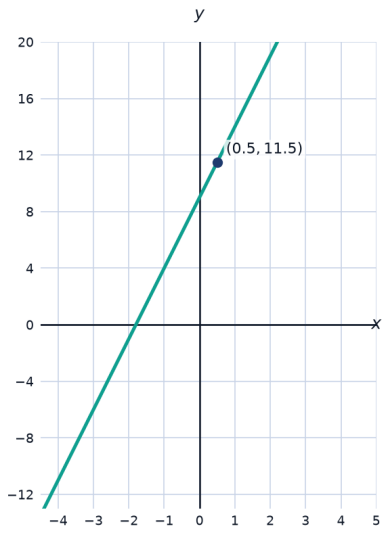
Quadratic Functions

ClipSAT Printable Practice Suite (clipsat.org) • Topic 2.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

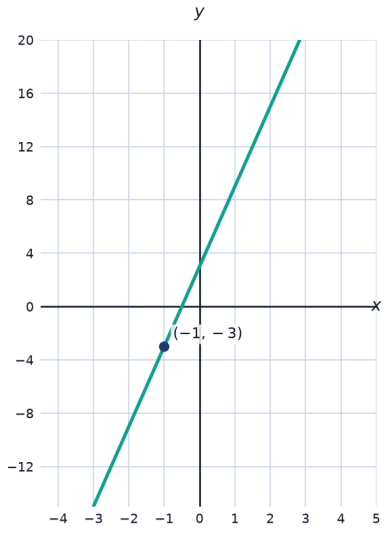
1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 5x + 9$ in solving multi-step equations, evaluate $f(5)$.



2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 6x + 3$ in literal equations and formulas, evaluate $f(2)$.



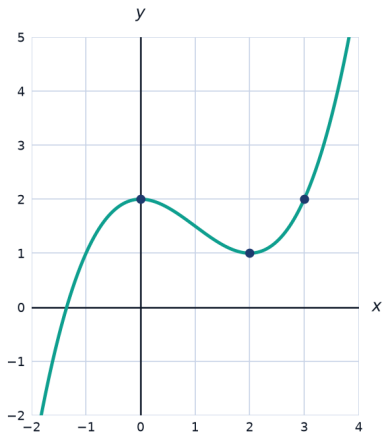
3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A data set recorded for literal equations and formulas consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean? Show your complete algebraic work.

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4 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 2x + 4$ in practical problems with equations, evaluate $f(2)$.



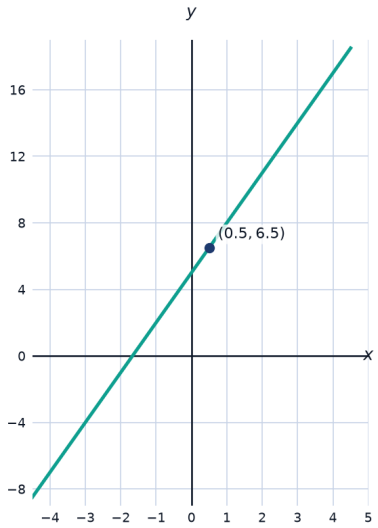
5 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A data set recorded for practical problems with equations consists of the four values $\{16, 20, 24, 30\}$.

What is the arithmetic mean? Show your complete algebraic work.

6 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

For the model $f(x) = 3x + 5$ in review: solving one-step inequalities, evaluate $f(2)$.



7 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A data set recorded for review: solving one-step inequalities consists of the four values $\{17, 21, 25, 31\}$.

What is the arithmetic mean? Show your complete algebraic work.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

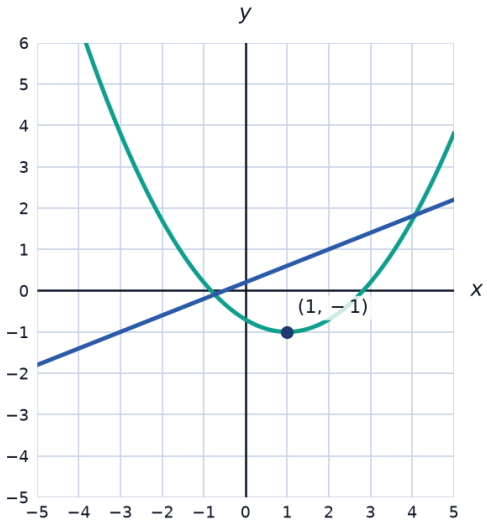
Find all solutions to $2x^2 - 6x + 5 = 0$ using the quadratic formula, and determine whether the roots are real or complex.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

[Quadratic Functions · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Factor the quadratic expression $6x^2 + 11x - 10$ completely.



11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Given $f(x) = \frac{2}{x-1}$ and $g(x) = \sqrt{x+5}$, find (a) the value of $(f \circ g)(4)$, and (b) the domain of $g(x)$.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A culture of bacteria doubles in population every 3 hours. If the initial count is 250 bacteria, write an exponential model $B(t)$ for the population after t hours, and find the population after 12 hours.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

[Exponential & Logarithmic Functions · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

[Functions & Transformations · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

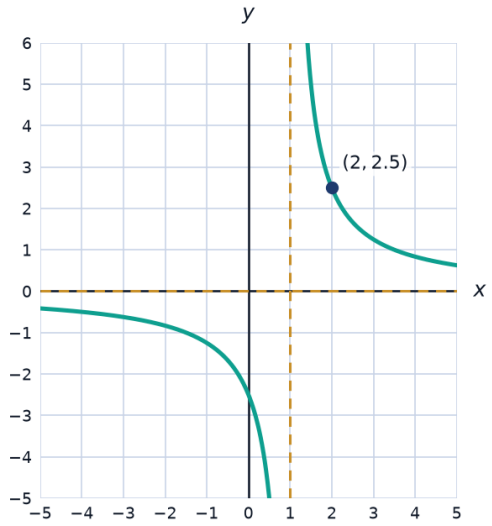
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15 **[Challenging]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)** • **Cross-Lesson: 7.01 Sequences & Series**

Simplify the algebraic expression $\frac{(2x^3y^{-2})^3}{(4x^{-1}y^4)^2}$ leaving only positive exponents.

16 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)** • **Cross-Lesson: 8.01 Factoring Polynomials**

Simplify and rationalize the denominator: $\frac{\sqrt{3}}{\sqrt{5}-\sqrt{2}}$.



17 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)** • **Cross-Lesson: 9.01 Radicals & Exponents**

Factor the polynomial completely by grouping: $2x^3 - 3x^2 - 8x + 12$.

18 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)** • **Cross-Lesson: 1.01 Linear Equations & Lines**

Solve the linear inequality $-3(2x - 4) \le 4x - 8$ and express the solution in interval notation.

19 **[Challenging]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)** • **Cross-Lesson: 10.01 Rational Expressions**

[Exponents & Polynomials · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

20 **[Challenging]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)** • **Cross-Lesson: 11.01 Statistics & Data Analysis**

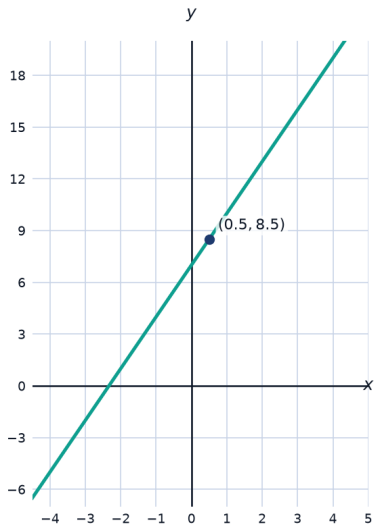
[Factoring Polynomials · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 7$ in review: solving one- and two- step equations, evaluate $f(2)$.



a. 13

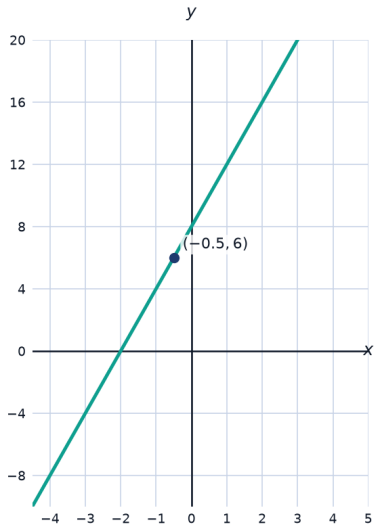
b. 16

c. 6

d. 17

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 4x + 8$ in properties of equality, evaluate $f(2)$.



a. 16

b. 20

c. 8

d. 20

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for review: solving one- and two- step equations consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5

d. 20.5

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for properties of equality consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5

d. 21.5

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The graph of the quadratic function $f(x) = ax^2 + bx + c$ passes through the points $(1, 5)$, $(-1, 1)$, and $(2, 10)$. Determine the values of a, b, and c.

a. $a = 2, b = -1, c = 4$

b. $a = 1, b = 2, c = 2$

c. $a = 3, b = -2, c = 4$

d. $a = 2, b = 1, c = 2$

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Which of the following equations represents a parabola that opens downwards?

a. $y = 2x^2 + 3x - 1$

b. $y = -x^2 + 5$

c. $y = x^2 - 4x$

d. $y = 3x^2$

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the quadratic function $f(x) = 2x^2 - 8x + 6$. What is the y-intercept of the graph of this function?

a. -8

b. 6

c. 2

d. -4

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28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

What is the larger root of $x^2 - 2x - 48 = 0$?

a. 9

b. 8

c. 7

d. -6

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find the x-coordinate of the vertex of $f(x) = -1 - x^2 + 3x$.

a. $\frac{3}{2}$

b. $\frac{5}{2}$

c. 3

d. $-\frac{3}{2}$

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Let $f(x) = e^{2x} - 5e^x + 6$. The minimum value of $f(x)$ occurs at a point $x = a$. If the equation $f(x) = k$ has exactly one real solution for x, find the value of k.

a. $k = 0$

b. $k = -1/4$

c. $k = 1/4$

d. $k = 1$

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In the coordinate plane shown, triangle $A'B'C'$ is obtained by first reflecting triangle ABC across the line $y = x$, and then translating the resulting image by h units horizontally and k units vertically. What is the ordered pair (h, k)?

a. $(-4, -2)$

b. $(-3, -3)$

c. $(-2, -4)$

d. $(4, 2)$

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32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A function $f : \mathbb{R} \to \mathbb{R}$ satisfies the following properties: 1. The graph of $y = f(x)$ is symmetric with respect to the line $x = 2$. 2. The graph of $y = f(x)$ is symmetric with respect to the line $x = 5$. 3. For all $x \in [2, 5]$, $f(x) = -(x - 3)^2 + 4$. Let $g(x) = f(x - 1) + 1$. How many real solutions does the equation $g(x) = \frac{1}{3}x$ have for $x > 0$?

a. 3

b. 4

c. 5

d. 6

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Which of the following is the completely factored form of the quadratic expression $x^2 - 5x - 14$?

a. $(x - 7)(x + 2)$

b. $(x + 7)(x - 2)$

c. $(x - 14)(x + 1)$

d. $(x - 7)(x - 2)$

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

An exponential function is given by $f(x) = a \cdot b^x$, where a and b are positive constants. The graph of $y = f(x)$ in the xy-plane passes through the points $(0, 4)$ and $(2, 36)$. What is the value of b?

a. 3

b. 6

c. 9

d. 16

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

The graph of the function $y = f(x)$ is transformed to produce the graph of $y = g(x)$. The graph of $y = g(x)$ is obtained by applying the following transformations to the graph of $y = f(x)$ in order: a vertical stretch by a factor of 3, followed by a horizontal shift of 4 units to the left, and finally a reflection in the x-axis. Given that $f(x) = x^2$, determine the equation of $y = g(x)$.

a. $g(x) = -(x + 4)^2 + 3$

b. $g(x) = -3(x + 4)^2$

c. $g(x) = -(x - 4)^2 + 3$

d. $g(x) = -3(x - 4)^2$

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36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Which of the following represents the exponential function $f(x) = b^x$ where the base b is greater than 1?

a. A graph that is always decreasing and passes through (0,1)

b. A graph that is always increasing and passes through (0,1)

c. A graph that is always decreasing and passes through (0,0)

d. A graph that is always increasing and passes through (0,0)

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Given the exponential function $f(x) = 5^x$, what is the value of $f(-2)$?

a. $\frac{1}{25}$

b. -10

c. 25

d. $-\frac{1}{10}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Linear Equations & Lines

The graph of the function $y = f(x)$ is shown below. Which of the following represents the graph of $y = f(x + 2)$?

a. Graph A

b. Graph B

c. Graph C

d. Graph D

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 10.01 Rational Expressions

Which of the following is the correct factored form of the quadratic expression $x^2 - 5x + 6$?

a. $(x - 2)(x - 3)$

b. $(x + 2)(x + 3)$

c. $(x - 1)(x - 6)$

d. $(x + 1)(x + 6)$

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 11.01 Statistics & Data Analysis

A function is defined by $f(x) = 2x + 3$. What is the value of $f(4)$?

a. 10

b. 11

c. 7

d. 14

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Exponents & Polynomials

A population starts at 300 and multiplies by 2 every year. What is the population after 2 years?

a. 1200

b. 2400

c. 600

d. 1500

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 4.01 Functions & Transformations

Which is the inverse of $y = e^x$?

a. $y = \frac{1}{e^x}$

b. $y = \ln x$

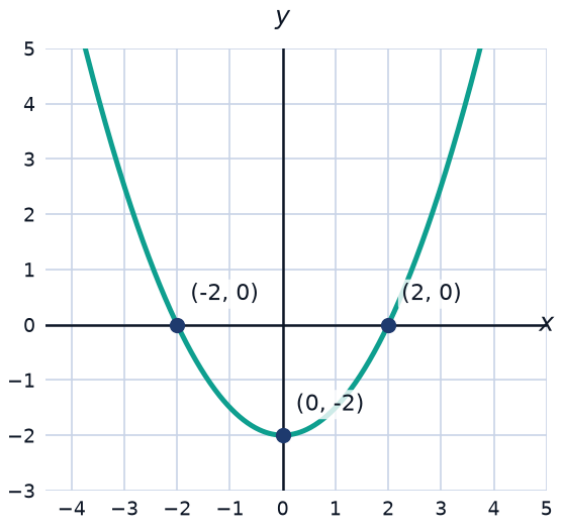
c. $y = \log x$

d. $y = e^{-x}$

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

The graph of the quadratic function $f(x) = \frac{1}{2}x^2 - 2$ is shown in the figure below. How many distinct real solutions does the equation $(f \circ f \circ f)(x) = 0$ have?



a. 3

b. 4

c. 6

d. 8

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Systems & Inequalities

The polynomial $P(x) = 2x^4 - 7x^3 + ax^2 + bx - 12$ has $x^2 - 4$ as a factor, where a and b are real constants. Which of the following is also a factor of $P(x)$?

a. $2x - 1$

b. $2x + 1$

c. $x + 3$

d. $2x - 3$

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 7.01 Sequences & Series

Consider the equation $a^{\sqrt{x}} \cdot b^{\sqrt{x}} = (ab)^{\sqrt{x}}$ and $a^{\sqrt{x}} \div b^{\sqrt{x}} = \left(\frac{a}{b}\right)^{\sqrt{x}}$ where $a, b > 0$ and $aeq1, beq1$. If $k = \sqrt{x}$, find the value of k that satisfies the equation $\left(\frac{3^{k^2+2k+1}}{2^{k^2+2k+1}}\right) \cdot \left(\frac{2^{2k+2}}{3^{2k+2}}\right) = \left(\frac{3}{2}\right)^{5k-1}$.

a. 3

b. 1/3

c. 1/5

d. 5

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